NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – IV : BIO-MASS & GEOTHERMAL ENERGY

1. Introduction to Biomass

Biomass = organic matter derived from plants and animals (wood, crop residue, animal dung, municipal & industrial organic waste, algae, energy crops) – stored form of **solar energy** through photosynthesis.

$$6CO_2 + 6H_2O + light \rightarrow C_6H_{12}O_6 + 6O_2$$

- Photosynthesis efficiency ≈ 1% (up to 3–5% for sugar cane).
- **Carbon neutral:** CO_2 released when burnt = CO_2 absorbed during growth.
- Sources: forest wood, agricultural residue (rice husk, bagasse, straw, coconut shells), cattle dung, urban sewage & MSW, aquatic plants (water hyacinth, algae), energy plantations (Jatropha, Pongamia, sugarcane, Subabul).
- India: ~ 500 million tonnes biomass/yr → ~ 18,000 MW potential; 12 million biogas plants possible.

❖Biomass conversion technologies

Route	Process	Product
Thermo-chemical	Direct combustion, pyrolysis, gasification, liquefaction	Heat, char, producer gas, bio-oil
Bio-chemical	Anaerobic digestion, fermentation	Biogas, ethanol
Chemical	Esterification (trans-esterification)	Biodiesel

2. Principles of Bio-Conversion

- Conversion of organic matter into usable fuels with the help of **micro-organisms (bacteria, enzymes)** under controlled conditions.
- **Aerobic digestion** (with oxygen): organic matter + O_2 → CO_2 + H_2O + heat + compost. Fast, gives **compost** and heat; no usable gas; used in aerobic composting & sewage activated sludge.
- **Anaerobic digestion (AD)** (without oxygen): organic matter → **biogas** (CH_4 + CO_2) + digested slurry (manure).

- Fermentation: sugars → ethanol + CO_2 by yeast: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$.

❖Anaerobic digestion – 3 stages

- Stage 1 – Hydrolysis (liquefaction): *hydrolytic bacteria* (acidogens) break down complex carbohydrates, proteins & fats into simple sugars, amino acids, fatty acids.
- Stage 2 – Acid formation (acidogenesis & acetogenesis): *acid-forming bacteria* convert these into volatile fatty acids (acetic, propionic), alcohols, H_2 and CO_2.
- Stage 3 – Methanogenesis: *methanogenic bacteria* (Methanobacterium, Methanosarcina) convert acetic acid, H_2/CO_2 into methane:

$$CH_3COOH \rightarrow CH_4 + CO_2 \; ; \; CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

❖Factors affecting biogas production

- Temperature: psychrophilic (< 20 °C), mesophilic (30–40 °C – best for village plants), thermophilic (50–60 °C).
- pH: 6.8 – 7.5 optimum (acid pH inhibits methanogens).
- C/N ratio: 20–30 : 1 optimum (cattle dung ≈ 24; straw ≈ 80 → mix with urine/ sewage).
- Total solids: 7–9% (dung : water = 1 : 1); retention (detention) time 30–50 days (mesophilic); loading rate 0.5–1.6 kg VS/m^3d.
- Agitation/ stirring, absence of toxic substances (antibiotics, detergents, heavy metals), seeding with active slurry.
- Particle size: smaller is better.

3. Biogas – Composition & Properties

Constituent	% by volume
Methane CH_4	55 – 65
Carbon dioxide CO_2	35 – 45
H_2S, N_2, H_2, CO	traces (< 1–2)

- Calorific value ≈ 20 – 25 MJ/m^3 (4700–5500 kcal/m^3) – vs. pure CH_4 36 MJ/m^3.

- Density ≈ 1.15 kg/m^3; ignition temperature 650–750 °C; flammability limit 5–15% in air; flame speed low (≈ 0.25 m/s); burns with clean blue flame; air needed ≈ 5.7 m^3 per m^3 gas.
- Octane number ≈ 110–130 → good for SI engines.
- Slurry after digestion = enriched manure (N, P, K retained, odour-free, weed-seed free).

4. Types of Biogas Digesters (Plants)

❖A. Batch type vs continuous type

- Batch: loaded once, sealed, emptied after gas stops (30–60 days) – used for straw/ MSW; gas production variable → needs several digesters.
- Continuous: fed daily; same volume removed; steady gas (village plants).
- One-stage vs two-stage, dry vs wet digestion.

❖B. KVIC floating drum type (India)

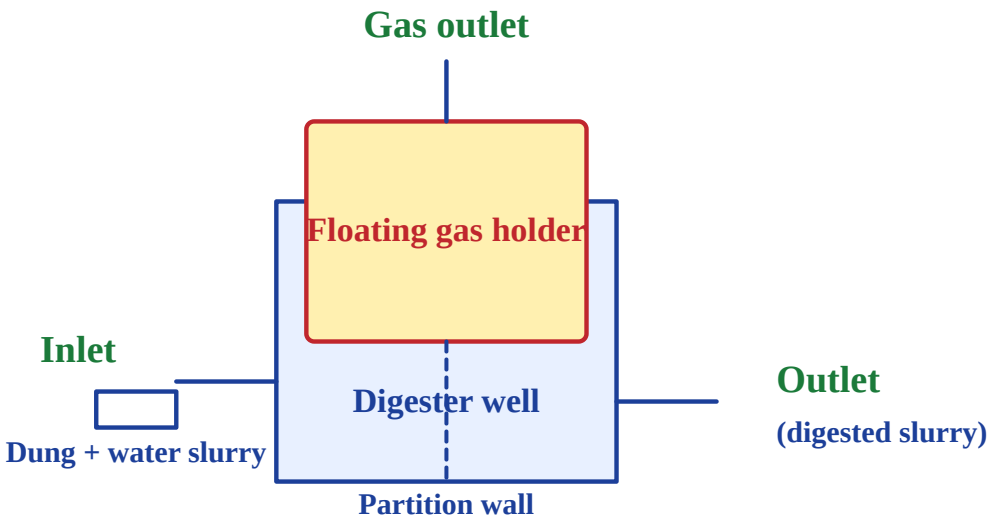


Fig 4.1 KVIC floating drum biogas plant

- Brick/ concrete digester well with central partition wall; floating steel drum gas holder moves up/down with gas storage – gives constant gas pressure (10–15 cm WG).
- Feed in through inlet pipe, slurry overflow at outlet; drum guided by central guide pipe.
- Advantages: constant pressure, volume of gas visible; easy to operate.
- Disadvantages: costly steel drum, corrosion (repaint yearly), higher maintenance, drum may stick in scum.

❖C. Fixed dome type (Chinese / Janata / Deenbandhu)

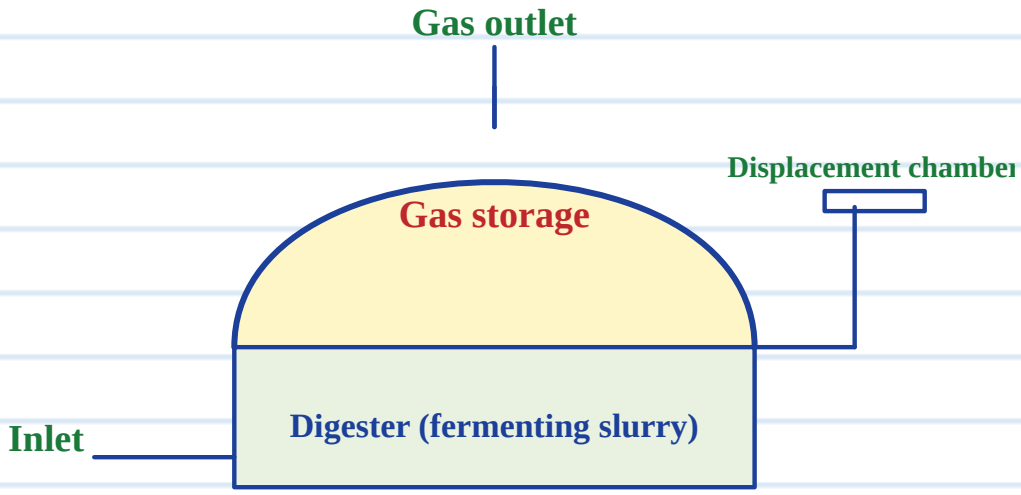


Fig 4.2 Deenbandhu / fixed dome (Chinese) biogas plant

- Underground masonry **dome** shaped digester with fixed gas holder on top; gas pressure builds as slurry is displaced into the **displacement chamber** (outlet tank) → **variable pressure** (up to 1 m WG).
- **Deenbandhu model:** spherical segments, low-cost, 1.0–2.5 m^3/day; ~ 30% cheaper than KVIC.
- **Advantages:** low cost, no moving part, long life (20 yr), underground → temperature stable, saves space.
- **Disadvantages:** gas pressure fluctuates, cracks in dome cause leakage, harder to repair, gas volume not visible.

❖D. Other types

- **Bag digester / balloon type** (PVC/ rubber bag – portable), **Plug-flow, UASB** (up-flow anaerobic sludge blanket) for industrial effluent, **Anaerobic filters, CSTR** (continuous stirred tank reactor), **Community & institutional plants, Biogas from MSW/ landfill.**

❖Comparison

Feature	KVIC (floating)	Deenbandhu (fixed dome)
Gas holder	Steel drum floats	Masonry dome, fixed
Gas pressure	Constant	Variable
Cost	High	Low (~ 30% less)
Life	8–15 yr (drum)	20+ yr
Maintenance	Painting needed	Very little
Gas leakage	Rare	Possible via cracks

5. Design & Gas Yield

- Gas yield: cattle dung ≈ 0.036 m^3/kg fresh dung (~ 0.25–0.3 m^3/kg dry VS); poultry droppings 0.07; pig dung 0.06; human excreta 0.07; water hyacinth 0.04; kitchen waste 0.1–0.2 m^3/kg.
- 1 cow/ buffalo gives ~ 10–15 kg dung/day → 0.36–0.5 m^3 gas/day.
- Daily dung input = (gas required) / (yield per kg); volume of digester = (slurry per day) × (retention time).

Digester volume $V_d = Q_{slurry} \times HRT$; Gas holder = 50–60% of daily gas

- Per-person gas for cooking ≈ 0.34 m^3/day (0.227 m^3 for lighting 1 lamp/ h).

❖Design example

> A family needs 2 m^3 gas per day. Dung needed = 2/0.036 = 55.6 kg/day (≈ 5 cattle) mixed with 55.6 L water → slurry 111 L/day. Retention 40 days → V_d = 0.111×40 = 4.4 m^3 (use 4.5 m^3 plant); gas holder ≈ 1.2 m^3.

❖Typical plant sizes (India): 1, 2, 3, 4, 6, 8, 10, 15, 20, 25 m^3/day.

6. Combustion Characteristics of Biogas

- Gas is a mixture of CH_4 + CO_2; CO_2 is incombustible diluent → lower flame temp (~ 800–850 °C) vs LPG (1900 °C).
- Chemical reaction: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + 35.8 MJ/m^3.
- Stoichiometric air: about 5.7 m^3 / m^3 (≈ 9.5 m^3 for pure methane).
- Flame speed is low (~ 0.25 m/s vs 0.45 m/s for LPG) → burner design must avoid flame blow-off; needs larger burner ports, low gas pressure, air-entrainment (primary air 60%) and orifice for high gas flow.
- Wobbe index is used to compare interchangeability of fuels; biogas ≈ 20–25 MJ/m^3.
- Purification: remove H_2S (iron oxide/ iron sponge → corrosion & SO_2), moisture (water trap), and CO_2 (water scrubbing, amine, PSA) → bio-methane (CBG) 95% CH_4 usable as vehicle fuel.
- Safety: gas leak is explosive at 5–15% in air; use ventilated room; do not use naked flame near the plant.

7. Utilisation for Cooking

- **Biogas stove:** simple burner with gas jet, air-mixing tube, burner head of 12–24 holes. Efficiency **55–60 %** (vs. wood chulha 10–15%, kerosene 40%).
- Gas consumption: 0.2–0.3 m^3/ person-day for cooking; boiling 1 L water ≈ 0.03 m^3.
- Gas pressure 75–100 mm WG required at burner; use pipes of 15–25 mm dia (GI/ PVC/ HDPE), valves, **water-drain traps**, pressure gauge.
- **Biogas lamps:** mantle lamps (0.1–0.15 m^3/h gives 40–60 W bulb light).
- **Other uses:** biogas **refrigerators**, **water heating**, **incubators**, **dual-fuel/ electricity**, **cogeneration** in dairy farms, CBG for transport.
- **Advantages over wood:** no smoke (saves women's health), no deforestation, saves time, hygienic kitchen.

8. Operation of I.C. Engines on Biogas

- Biogas is an **excellent fuel for SI engines** (high octane ~ 110–130) – can run **with petrol** (dual fuel up to 100% biogas) or fully on gas (modified carburettor/ gas mixer).
- **Diesel engines** cannot run on biogas alone (self-ignition temp 650 °C vs 250 °C for diesel) → **dual-fuel mode: pilot diesel injection (10–20%)** ignites the biogas–air mixture; saves 60–80% diesel.
- **Modifications:** gas–air mixer/ venturi in the inlet; reduce compression ratio for SI (≤ 12:1), or increase CR for gas-only SI engines; advance ignition timing (slow burning) ; corrosion-resistant materials; H_2S removal (it corrodes valves, oil).
- **Power derating:** 10–20% less power than petrol because gas displaces air (lower volumetric efficiency).
- **Specific consumption:** about **0.5–0.6 m^3 per kWh** (BHP-h ≈ 0.45 m^3); 1 m^3 gas ≈ 1.5 kWh electricity ≈ 0.6 L diesel ≈ 0.75 L petrol ≈ 0.4 kg LPG.
- **Uses:** **village power generation (genset 5–50 kW)**, water pumping, flour mills, chaff cutters.
- Engine exhaust heat can heat the digester (cogeneration).

❖Engine efficiency

$$\eta_{th} = P / (\dot{m}_{gas} \times CV) \approx 25 - 30\ \%\ \text{for gas engine}$$

9. Economic Aspects of Biogas

- **Capital cost:** 2 m^3 family plant ≈ ₹ 20,000–30,000 (subsidy through MNRE / NNBOMP ~ ₹ 7,000–12,000, extra for toilet-linked).
- **Benefits per year (2–3 m^3 plant):** saves LPG (≈ 6–8 cylinders), fire-wood (~ 3 tonnes), gives manure (~ 12 tonnes slurry; replace chemical fertilizer worth ₹ 5000+).
- **Payback period:** 3–6 years; life 15–20 years.
- **Social benefits:** clean air, women/ child time saving, sanitation, rural jobs, reduced GHG (methane capture), less deforestation.

Annual saving S ; investment C ; Simple payback = C / S (years)

❖Example

> Plant cost ₹ 25,000 (after subsidy ₹ 15,000). Annual savings: LPG 7 × ₹ 900 = 6300; fuel-wood ₹ 3000; manure ₹ 4000; total = ₹ 13,300/yr → payback = 15000/13300 ≈ 1.1 yr. Without subsidy = 1.9 yr.

- **Operating cost:** negligible – only labour for dung feeding, minor repairs.

10. Other Biomass Conversion (short notes)

- **Direct combustion:** wood/ husk boilers (efficiency 70–80%), bagasse cogeneration in sugar mills.
- **Pyrolysis:** heating biomass at 400–600 °C in absence of air → char + bio-oil + gas.
- **Gasification:** partial combustion (air-deficient, 800–1000 °C) → producer gas ($CO + H_2 + CH_4$, 4–6 MJ/m^3); downdraft gasifier for engines (1 kWh ≈ 1.2 kg wood).
- **Ethanol:** fermentation of molasses/ cane juice/ grain; blended in petrol (E20 target in India).
- **Biodiesel:** trans-esterification of vegetable oils (Jatropha/ Pongamia) with methanol.
- **Briquetting / pelletisation** of agro-waste.

II. GEOTHERMAL ENERGY – Introduction & Resources

> **Geothermal energy** = heat from inside the earth (Greek: geo = earth, therme = heat), originating from **radioactive decay** (U, Th, K) in the crust and **primordial heat** of the core.

- **Earth structure:** crust (~ 30–35 km), mantle, outer core (liquid), inner core (solid, ~ 5000–6000 °C).
- **Geothermal gradient:** average **25–30 °C per km** of depth; heat flux ≈ 0.06 W/m^2. Higher near tectonic plate boundaries, volcanoes, hot springs, geysers.
- Total heat content is huge; practical use limited to places with **high gradient** and a **fluid reservoir**.
- **Advantages:** base-load power (capacity factor 70–90%), clean, small land footprint, independent of weather, cheap running cost.
- **Disadvantages:** location-specific, high drilling cost, release of H_2S/ CO_2/ boron/ arsenic, scaling/ corrosion in pipes, possible land subsidence/ induced seismicity, need to re-inject fluid.

❖Classification of resources

1. **Hydrothermal convective systems**
   - **Vapour-dominated (dry steam):** superheated steam 180–240 °C (e.g., The Geysers – USA, Larderello – Italy).
   - **Liquid-dominated (wet steam / hot water):** water at 150–350 °C under pressure; flashes to steam when pressure drops (Wairakei – NZ, Cerro Prieto).
2. **Hot dry rock (HDR) / Enhanced Geothermal System (EGS):** hot impermeable rock at 3–5 km; water injected through one well, fractured rock heats it, returned via a production well.
3. **Geo-pressured reservoirs:** hot brine at high pressure with dissolved methane (Gulf of Mexico).
4. **Magma (molten rock) resources:** 600–1200 °C – future.
5. **Normal geothermal gradient (low-temp)** – used by **ground-source heat pumps**.

❖Resource temperature

Type	Temp	Use
High	> 150 °C	Electricity
Medium	90–150 °C	Binary power, industry
Low	< 90 °C	Space heating, spas, drying, greenhouses, heat pumps

12. Types of Wells

- **Production wells** – bring steam/ hot water up; depth 0.5–3 km (vertical, directional or slant drilled by rotary rigs with mud circulation).
- **Injection (re-injection) wells** – return spent fluid to reservoir → sustain pressure and avoid pollution/ subsidence.
- **Exploration (test) wells / slim holes** – to measure temperature, permeability, fluid chemistry.
- **Based on fluid: dry steam wells, wet (two-phase) wells, hot water wells; artesian** (self-flowing) vs **pumped** (down-hole pump).
- **Casing & cementing** – multiple casing strings protect against collapse and corrosion; **wellhead** valves control flow; **separator** removes water from steam.
- **Doublet system** (production + injection) is common for district heating and EGS.

13. Methods of Harnessing Geothermal Energy

❖(i) Dry-steam power plant

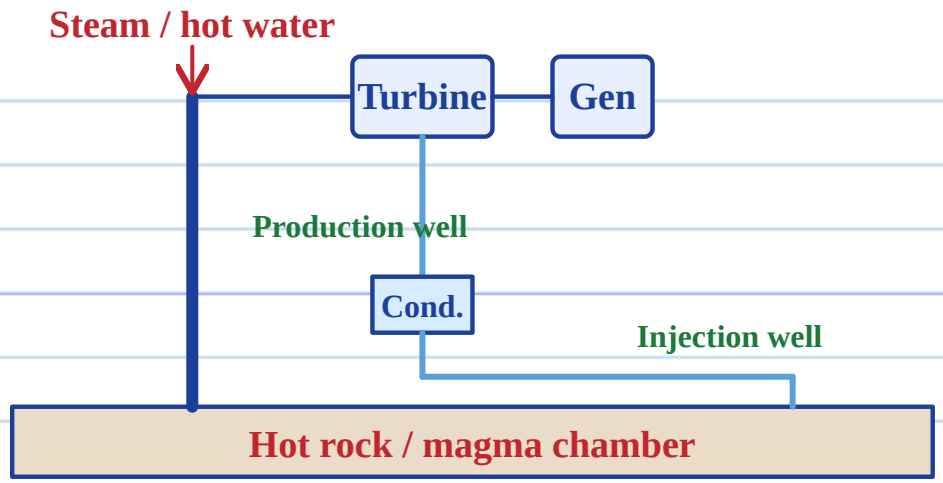


Fig 4.3 Dry-steam geothermal power plant

- Steam directly from production well (> 180 °C, 4–8 bar) → turbine → generator → condenser (cooling tower) → condensate re-injected. Simple, cheapest; e.g. The Geysers (≈ 1500 MW).

❖(ii) Flash-steam plant (single or double flash)

- High-pressure hot water (> 180 °C) is **flashed** in a separator (pressure drop) → steam goes to turbine; remaining brine re-injected or flashed a second time at lower pressure for more steam (**double flash** +15–25% power). Most common type (60% of world capacity).

❖(iii) Binary-cycle plant (ORC / Kalina)

- Geothermal water (100–180 °C) passes through **heat exchanger** to vaporise a **low-boiling working fluid** (isobutane, pentane, R-134a) in a closed **organic Rankine cycle**; geofluid never contacts the turbine and is fully re-injected → **zero emission**; suits low-temperature resources.
- Efficiency ~ 8–12%.

❖(iv) Hot dry rock (EGS)

- Water pumped down **injection well** → fracture network (hydraulic stimulation) → heated by rock → returns via **production well** → heat exchanger → power/ heat.

❖(v) Direct use (non-electric)

- **District & space heating** (Iceland – 90% homes), greenhouses, aquaculture, food drying, industrial process heat (paper, milk), spas/ balneology, snow melting, **ground-source heat pumps** (COP 3–5).

❖Thermodynamic notes

- Heat extraction: $Q = \dot{m}\,(h_1 - h_2)$; power for dry steam: $W = \dot{m}_s\,(h_{in} - h_{out})\,\eta_t$.
- Low source temperature ⇒ low efficiency (Carnot limit $\eta = 1 - T_c/T_h$; e.g. 150 °C source, 30 °C sink → 28%, actual ≈ 10–15%).
- Dryness fraction after flashing $x = (h_{f,in} - h_{f,sep}) / h_{fg,sep}$.

❖Flash example

> Well fluid 200 °C saturated liquid (h = 852 kJ/kg) flashed to 5 bar (h_f = 640, h_{fg} = 2109 kJ/kg): $x = (852 - 640)/2109 = 0.10$ → 10% of the mass becomes steam. For 100 kg/s well flow → 10 kg/s steam → with Δh = 700 kJ/kg across turbine, $P \approx 7$ MW (ideal).

14. Geothermal Potential in India

- ~ 340 hot spring locations, total potential estimated ≈ 10,600 MW (GSI/MNRE); no commercial geothermal power plant yet (pilot projects).
- Geothermal provinces:

Province	Examples
Himalayan	Puga & Chumathang (Ladakh) – 260 °C boiling (~ 100+ MW), Manikaran (H.P., 94 °C), Jwalamukhi
Sohana (Haryana)	Sohana
West Coast	Unkeshwar, Ratnagiri (Maharashtra), Tuwa (Gujarat)
Cambay graben	Cambay (Gujarat)
Son–Narmada–Tapi (SONATA)	Tattapani (Chhattisgarh), Anhoni (M.P.)
Godavari basin	Bhadrachalam, Manuguru (Telangana)
Mahanadi	Odisha – Atri, Deulajhari
Andaman & Nicobar	Barren Island (volcano), Narcondam

- Pilot: Manikaran 5 kW binary plant, Puga 1–2 MW project by ONGC–Iceland; Tattapani 300 kW.
- Challenges: low-grade resources, high cost, lack of data, environmental & land issues.

15. Solved Problems & Key Points

❖Problem 1 – Gas for cooking

A family of 5 needs 0.34 m^3/person/day = 1.7 m^3 gas/day. Dung = 1.7/0.036 = 47 kg/day (4 cows).

❖Problem 2 – Methane energy

100 m^3/day biogas at 22 MJ/m^3 = 2200 MJ/day = 611 kWh_{th}/day; with η = 28% → 171 kWh_e/day ≈ 7 kW average.

❖Problem 3 – Geothermal gradient

Gradient 30 °C/km, surface 25 °C: at 3 km → 25 + 90 = 115 °C; to reach 150 °C need 4.2 km.

❖Problem 4 – Carnot limit

Source 180 °C (453 K), sink 30 °C (303 K): $\eta = 1 - 303/453 = 33\ \%$ (practical 10–15%).

Key: Biogas = 55–65% CH_4, CV ≈ 22 MJ/m^3, mesophilic 35 °C, pH 7, C/N 25

16. Important Questions

- Explain the principle of bio-conversion; anaerobic vs aerobic digestion – three stages.
- With neat sketches explain KVIC floating drum and Deenbandhu fixed dome plants.
- Factors affecting gas yield; composition and combustion characteristics of biogas.
- How can biogas be used for cooking and to run I.C. engines? Economic aspects.
- Write about geothermal resources, types of wells and methods of harnessing.
- Discuss geothermal potential in India.